14-Process Workflow Diagram

AIoT-Based Smart Sheet-Metal Panel Fabrication System

P01–P09 | MANUFACTURING ENGINEERING

Customer Requirements → Component Library → Internal Arrangement → 3D CAD Design → BOM & Quotation → Flat Pattern → Laser Nesting → Cutting & Bending → Dimensional Verification, Painting & Assembly

P10–P11 | DIGITAL ENGINEERING & AIoT SUPPORT

P10: AIoT System Implementation Guide • P11: IoT Job-Tracking Dashboard

P12–P14 | INSPECTION & QUALITY ASSURANCE

P12: Camera-Based Dimensional Inspection → P13: Computer-Vision Feature Detection & Dimension Comparison → P14: Final Fabricated Enclosure & Quality Inspection

Manufacturing engineering sequence

P01 · Customer Requirement Collection Capture dimensions, material, rating, quantity and customer references.	**P02 · Component Identification & Digital Library** Identify, measure and catalogue components and mounting requirements.	**P03 · AI-Assisted Internal Component Arrangement** Prepare a proposed internal component layout for engineering review.
P04 · Final 3D CAD Enclosure Design Develop and review the enclosure CAD model and design features.	**P05 · CAD-to-Manufacturing Documentation, BOM & Quotation** Prepare manufacturing documentation, BOM and quotation inputs.	**P06 · Sheet-Metal Flat-Pattern Development & Validation** Develop and validate panel flat patterns and bend allowances.
P07 · Optimized Laser-Cutting Nesting Arrange validated part profiles on sheet stock for cutting preparation.	**P08 · Sheet-Metal Cutting, Bending & Component Fabrication** Perform the planned fabrication operations.	**P09 · Dimensional Verification, Painting & Complete Enclosure Assembly** Verify dimensions, complete finishing and assemble the enclosure.

Digital engineering and AIoT support

P10 · AIoT System Implementation Guide Defines proposed architecture, data flow, interfaces and integration boundaries.	**P11 · IoT Job-Tracking Dashboard** Represents job states and production events; live telemetry requires actual integration.

Inspection and quality assurance

P12 · Camera-Based Dimensional Inspection Capture images, calibrate the setup and extract measurements.	**P13 · Computer-Vision Feature Detection & Dimension Comparison** Compare detected features and dimensions against approved references.	**P14 · Final Fabricated Enclosure & Quality Inspection** Review evidence, resolve deviations and record the authorized quality decision.

Workflow interpretation: P01–P09 describe the manufacturing engineering sequence. P10–P11 are supporting digital/AIoT systems and do not represent additional physical manufacturing operations. P12–P14 form the inspection and quality-assurance path; inspection evidence and authorized review are required before a real final release. This diagram is a project overview, not a machine-control instruction.